Implication provability facts

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs ax:lnot2, ax:lif1.

Editorial note. The label prop:provability-lif names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Negation and implication axiom schemes. Same unit, ax:lnot2 / ax:lif1 (propositional axiom group). Item 2: each one-step derivation is an instance of the named scheme followed by the deduction theorem.

Source reference. Original reference ax-lif1: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 486–486. Prerequisite pointer invoked at proof line(s) 1085; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lnot2: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 489–489. Prerequisite pointer invoked at proof line(s) 1085; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *1.* $A, A \to B \vdash B$.

2. Both $\neg A \vdash A \to B$ *and* $B \vdash A \to B$.

Proof. 1. We can derive:

1.	A	Hyp
2.	$A \to B$	Hyp
3.	B	1, 2, MP

2. By ax-lnot2 and ax-lif1 and the deduction theorem, respectively.

□